

22 May 2018 v1

CS328A Signal Analyzer Description



Summary

The Cleverscope CS328A Signal Analyzer is a PC hosted system with two high resolution analog inputs, 8 digital inputs, and an isolated 65 MHz signal generator for the measurement and stimulus of a Unit Under Test (UUT).

The CS328A Cleverscope Acquisition Unit (CAU) pictured above uses either USB 2.0 or 100M Ethernet to communicate with the PC. The isolated Ethernet means the CAU can be in the lab or a test facility, while being controlled from a remote location.

Because we only transfer the data required for display we have a fast update rate and a high performance PC is not required unless high speed streaming is a requirement.

The CS328A Signal Analyzer PC application includes a rich set of tools:

- **Frequency Response Analyzer (FRA)**
Used to measure the amplitude, power, gain/phase, impedance, capacitance, inductance of the UUT over frequency. If the UUT is a live Power Supply Unit (PSU), the FRA can measure Power Supply Rejection Ratio, Input Impedance, Output Impedance, and PSU gain/phase over frequency.
- **Spectrum Analyzer**
Used to measure the wideband amplitude or power spectrum of a signal, and includes tools to measure harmonic frequencies and amplitudes, Sinad, and total harmonic distortion.
- **Time Analyzer - Oscilloscope**
Used to measure the UUT amplitude/time response to events and stimulus signals. A dual independent mixed signal trigger system finds hard to capture signals. The deep sample buffer and dual big picture/tracking zoom windows can be used to look at a 40 msec wide signal while observing 10ns resolution features at the same time, in real time. The Time Analyzer includes a tool to measure DC, RMS, Pk-Pk, Std Dev, Period, Fundamental Frequency and amplitude, and a variety of pulse statistics.
- **Maths Analyzer**
Used to do complex multi-line maths on the signal to derive real world values such as power, energy, and charge with live export to Matlab and Excel.

- **Streaming Signal Analyzer**

Capture long duration signals to check for intermittent events or the complete sequence of events at sampling rates of 3 Msamples/sec for over 100,000 seconds , and then quickly look for features and drill down into them.

- **Protocol Analyzer**

The Protocol Analyzer decodes UART, SPI, I2C and logic bus operations. It can use the full buffer memory to decode long messages, and automatically save them to a time stamped record which can easily be pasted into Excel. Messages are shown on the time graphs, with the decoded value placed at the start of a character, or concatenated with other characters if time scale is coarse.

Management functions

Besides the analyzer functions, the Cleverscope application makes it easy to understand, re-use and document signals being analyzed with these tools:

- Copy a graph or signal information, and paste into Word or Excel with Ctrl C and Ctrl V.
- Set real world names and units for graph axis and displayed units such as *T1 Current* and *Amp*
- Annotate the graphs to highlight salient features
- Add up to 50 reference waveforms for comparison between variations in UUT setup.
- Save the complete setup including waveforms, display and graph positions, annotations, names and units to disk. The resulting file can be emailed for others to look at, or re-opened as an exact-same setup. A most recently used file list makes it easy to open any one of the last 10 used saves.
- Automated save to disk in response to an event for overnight testing.

OEM embedded usage

Cleverscope provides a DLL or Labview VI so you can embed the CAU into your own test system. We provide worked examples in Microsoft Visual C++, C#, Basic, Delphi, Borland C++ and Labview. The DLL includes signal analysis functions such as spectrum, and all the derived measurements offered in the Cleverscope application. In addition it can be used for controlling up to 32 connected CAU's, and doing sequence captures of up to 3000 rapid fire frame captures and playbacks. Examples for this are included.

Lifetime Updates

Cleverscope provides free lifetime updates which include application software and CAU firmware to add new features. A simple tool is used to update the CAU in the field. Download the latest update from our website at any time.

Example Videos and Website

Cleverscope hosts a number of explanatory videos on our [Youtube](https://www.youtube.com/Cleverscope/videos) site at [youtube.com/Cleverscope/videos](https://www.youtube.com/Cleverscope/videos)

In addition visit the [Using Cleverscope](https://www.cleverscope.com/capabilities/) page at [cleverscope.com/capabilities/](https://www.cleverscope.com/capabilities/)

Cleverscope will expand the videos and explanatory material over time, so check back every now and then!

Conclusion

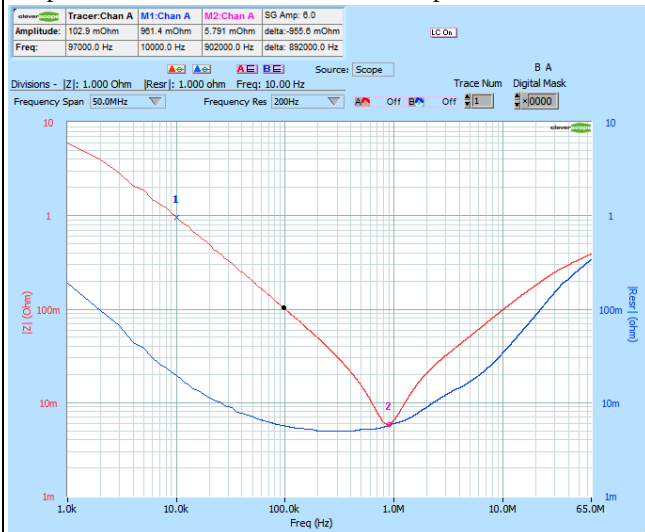
The Cleverscope Signal Analyzer is a comprehensive kit of tools to make the very most of the information you can extract from the UUTs operation, as the following pages will show. You can save all this information for later, as Murphy will inevitably make sure you were not paying attention!

Measurements

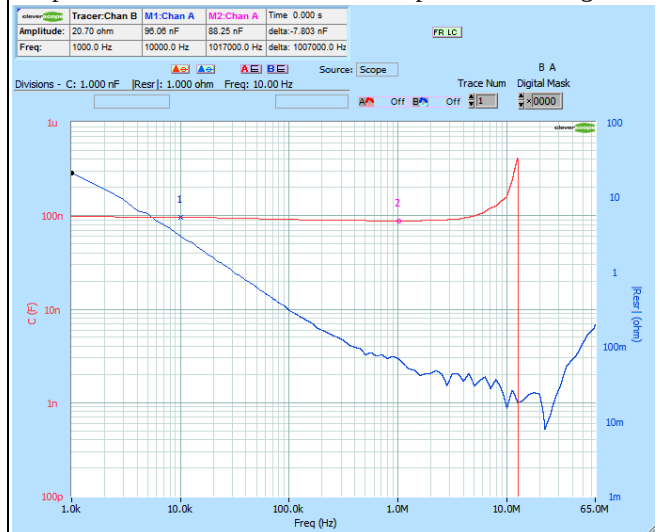
Frequency Response Analysis Functions (FRA)

The Frequency Response Analysis (FRA) system uses the isolated signal generator to provide stimulus for component, system or power supply measurements. Here are a collection of measurements made using the FRA system (zoom on the PDF to see the detail):

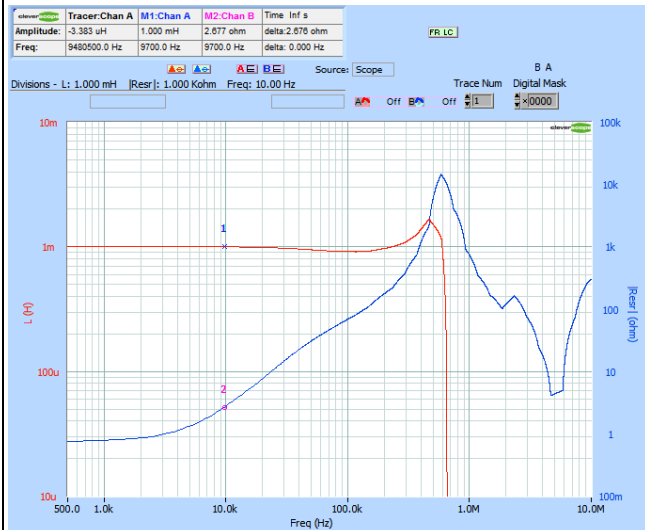
Impedance, Resr of a 22 μ F ceramic capacitor



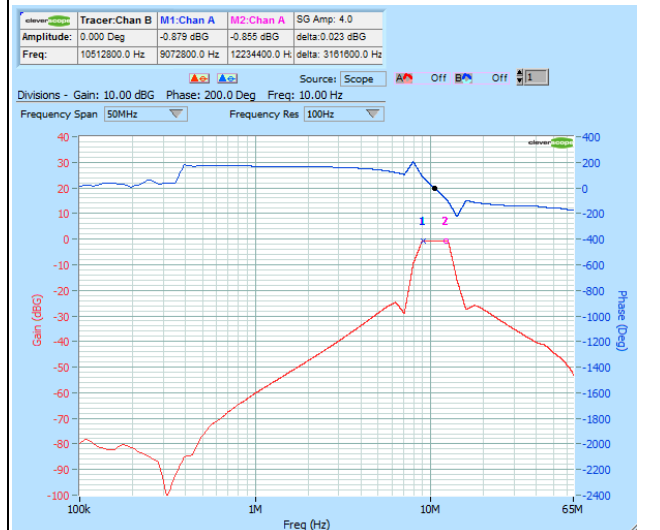
Capacitance, Resr of 100 nF 0603 capacitor, showing SRF



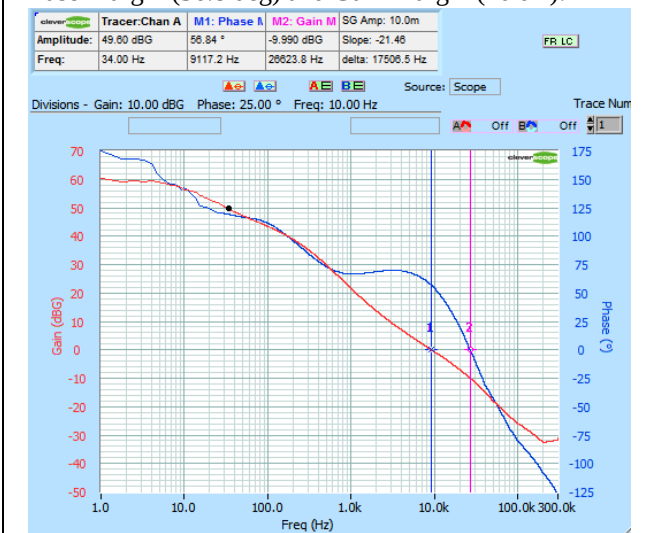
Inductance of 1 mH air cored wound audio inductor



Gain/Phase of a 10.7 MHz filter:



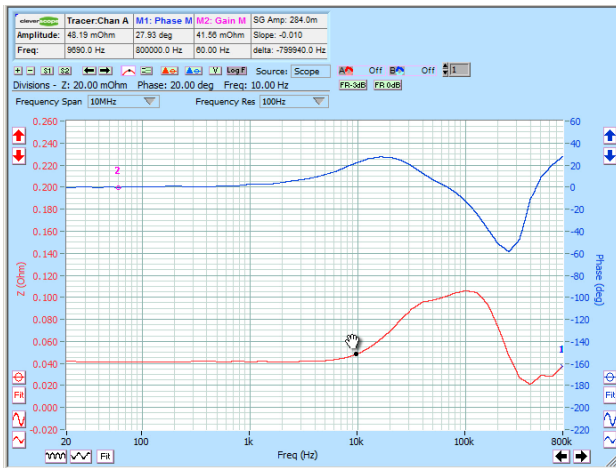
PSU Gain Phase of switching PSU TPS55340 showing Phase Margin (56.8 deg) and Gain Margin (20 dB):



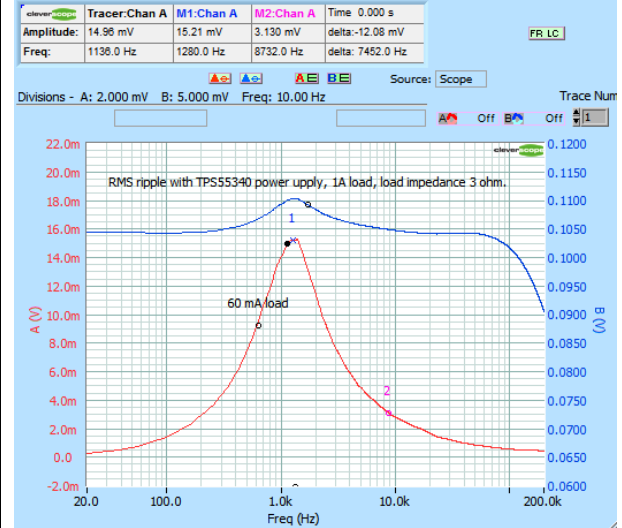
TLV70433 PSRR using CS1070 (1 A 50 MHz op-amp):



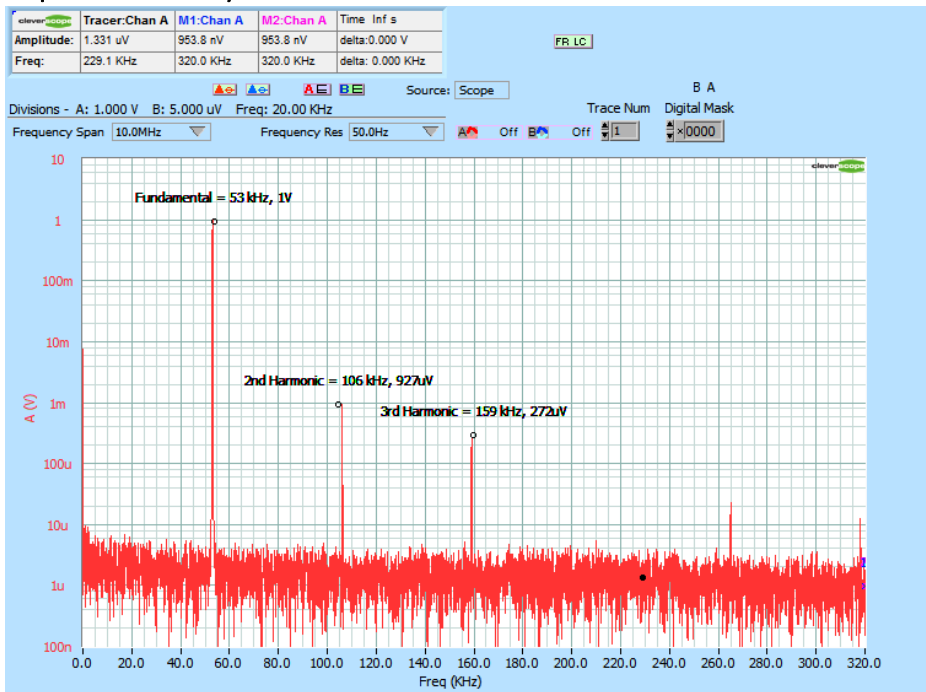
Powered LTC3589 Output Impedance using CS1070:



PSU Shunt Impedance and supply ripple TPS55340:



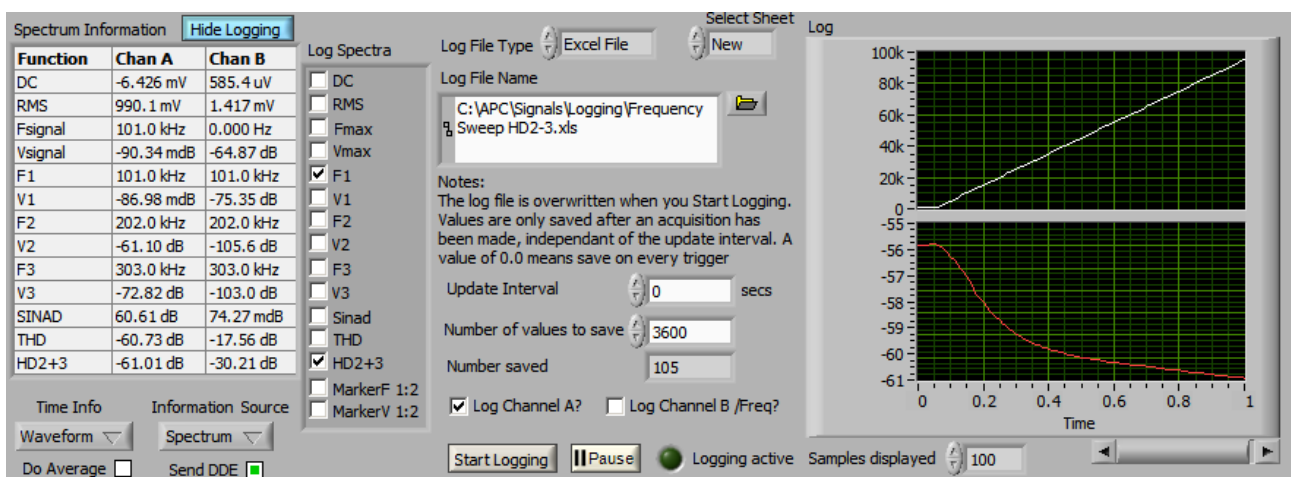
Spectrum Analyzer



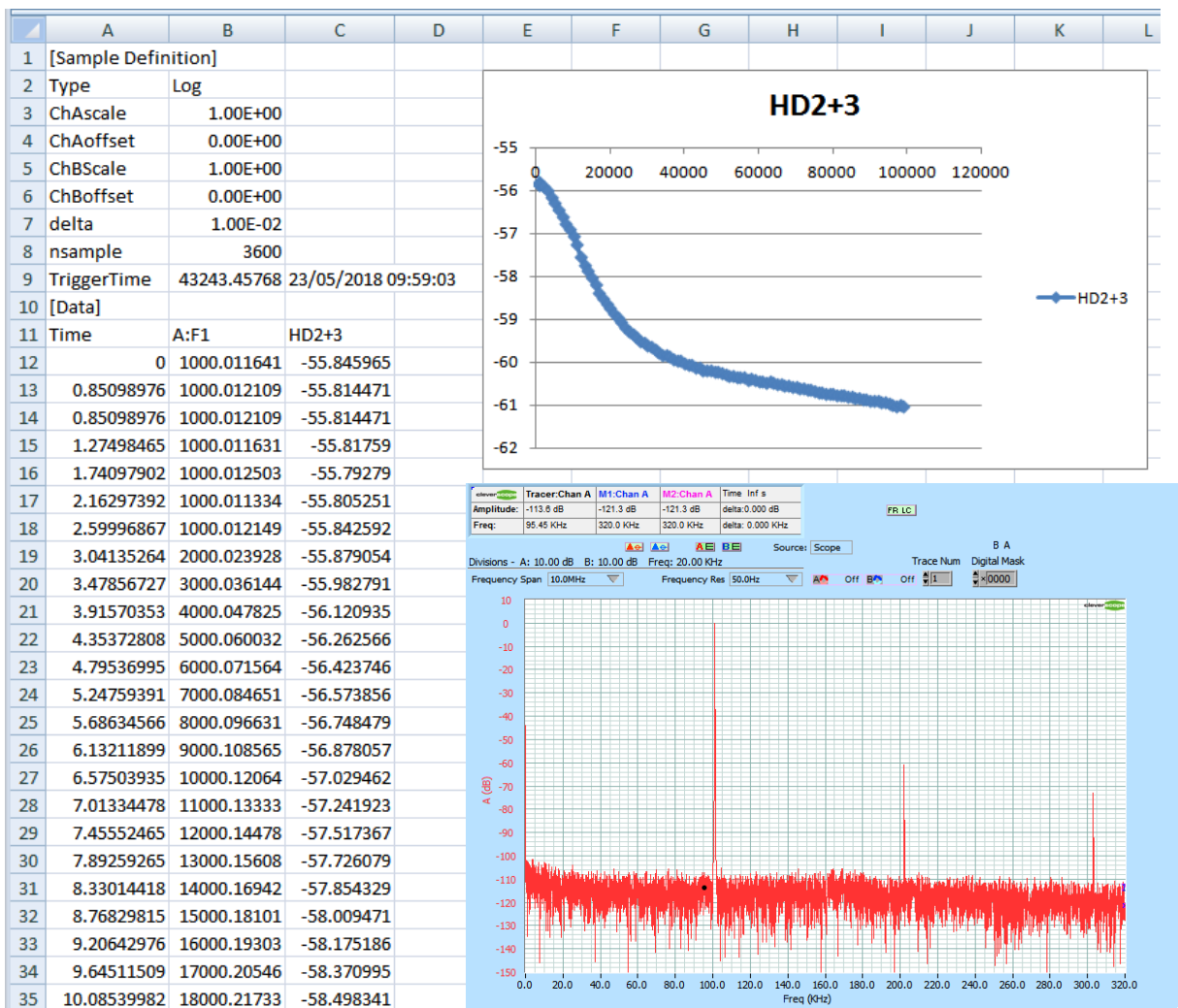
The Spectrum Analyzer displays in Volts or Power over Frequency. The amplitude axis can be in dB or V/W and can be formatted linear or log. The Frequency axis can also be formatted linear or log. Frequency Span and Resolution are selected from drop down menus. Here a 53 kHz 1V sine wave is displayed with log amplitude in V. The noise floor is below 10uV.

Note the annotations showing harmonics.

The **Signal Information** displays information derived from the Spectrum Display, and when expanded, used to log information to a text or excel file. Here the spectrum display was changed to dB, and we logged the RMS sum of Harmonic Distortion for the second and third harmonics vs frequency as the signal generator swept from 10 kHz to 100 kHz:



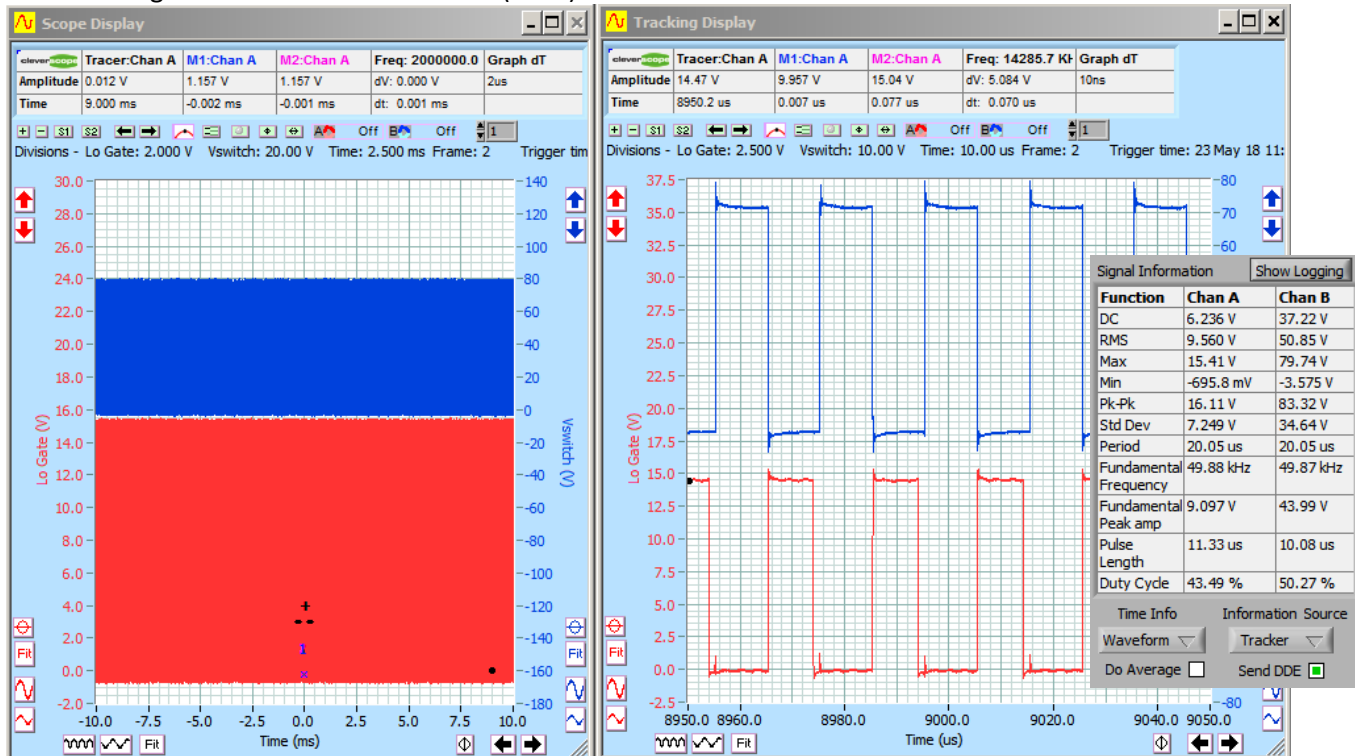
The Excel Spreadsheet is live updated and can be used for further analysis. Here we see the first few captures and the resulting HD2+3 vs Frequency graph. Interestingly HD2+3 actually gets better as frequency increases for the signal generator.



Inset is the dB response at the final 100 kHz frequency is shown. The noise floor is below 100 dBV.

Time Analyzer - Oscilloscope

The Time Analyzer - Oscilloscope uses the large sample buffer to give the big picture, and the fine detail at the same time. Here we have captured 20ms of the low side gate drive and half bridge switch node of a full bridge loaded with a 150 pF cap and small series inductance. Axis are labeled appropriately. Signal Information gives the waveform statistics (inset).

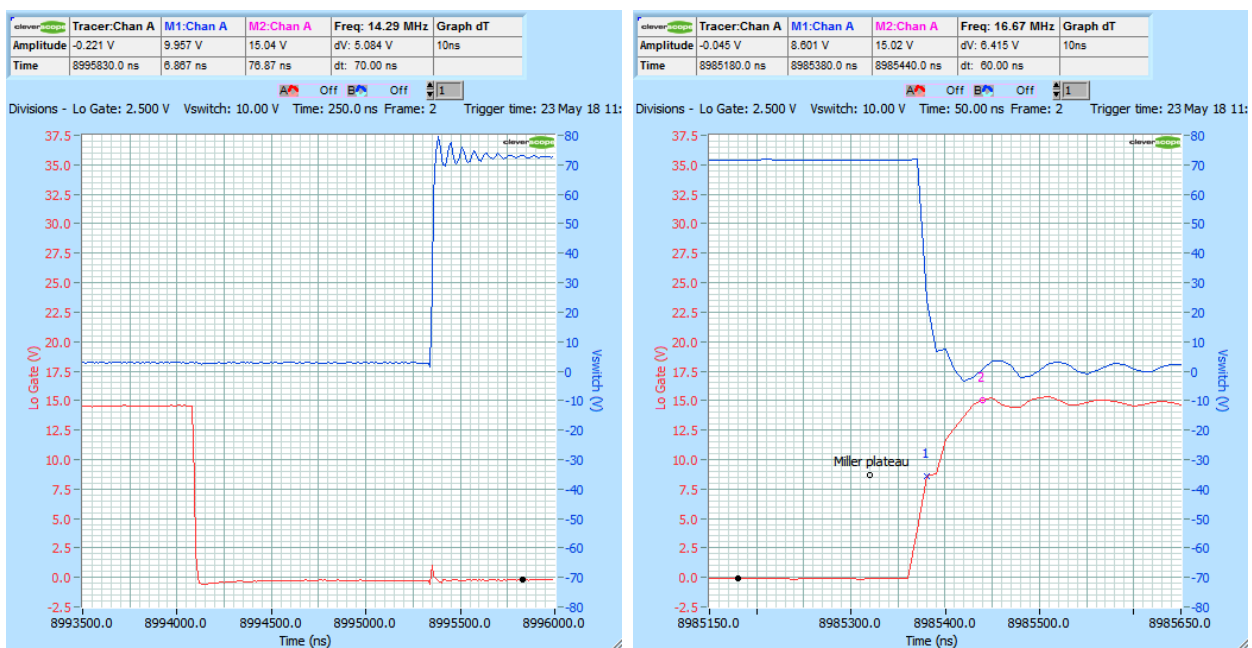


The Scope Display is the big picture view. The signal is scaled and offset to match the axis values, and presented to the ADC. Time zero is the trigger point.

The Tracking Display tracks the Tracer position on the Scope Display. Here it is located at 9000 us (see the black blob on the Scope Display). We can easily see the zoomed in 100 kHz gate drive and switch signals. The graph has a resolution of 10 ns (Graph dT). We can see Miller induced pulses in The gate Drive signal where the other Half bridge is switching, and the resonance of the L-C load.

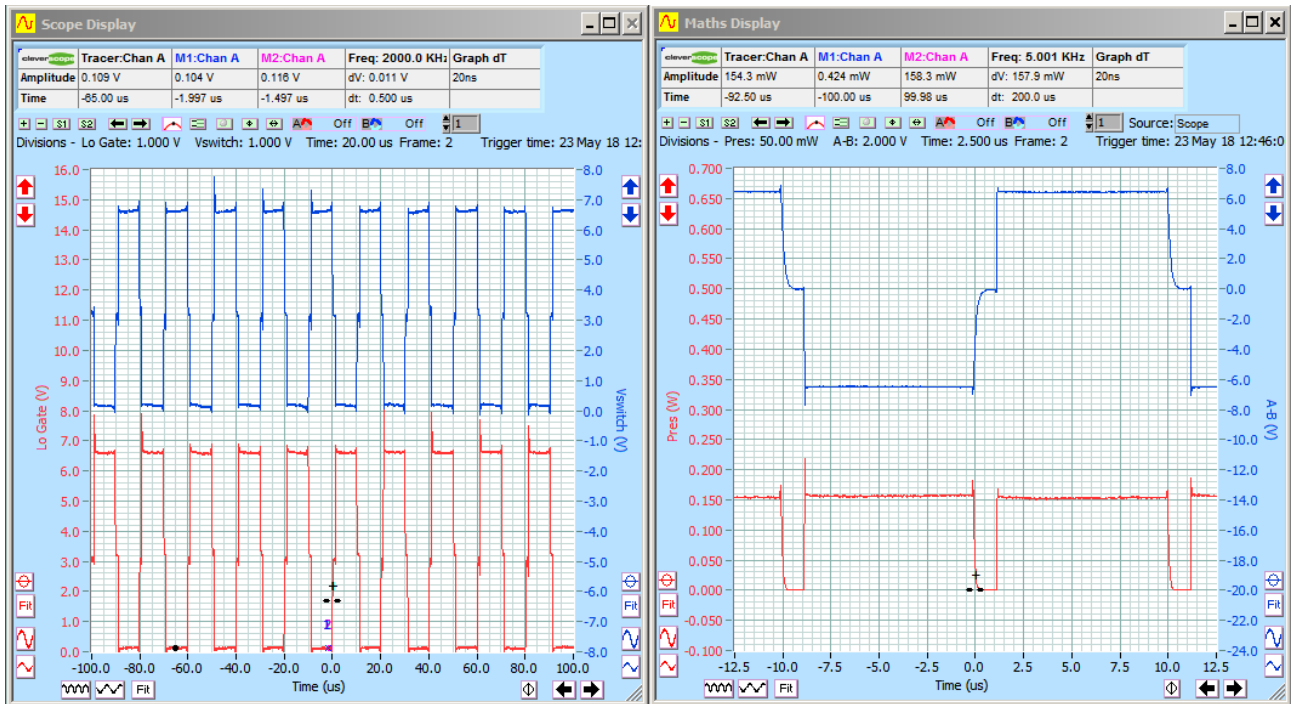
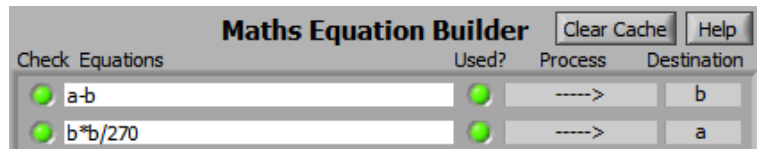
Zooming in further (250 ns/div) we can see a portion of a cycle with the tank oscillation quite visible.

..and here (at 50 ns/div) is the rising edge of the gate drive signal, with the Miller plateau clearly visible at 8.6 V, as the lower switch turns on.



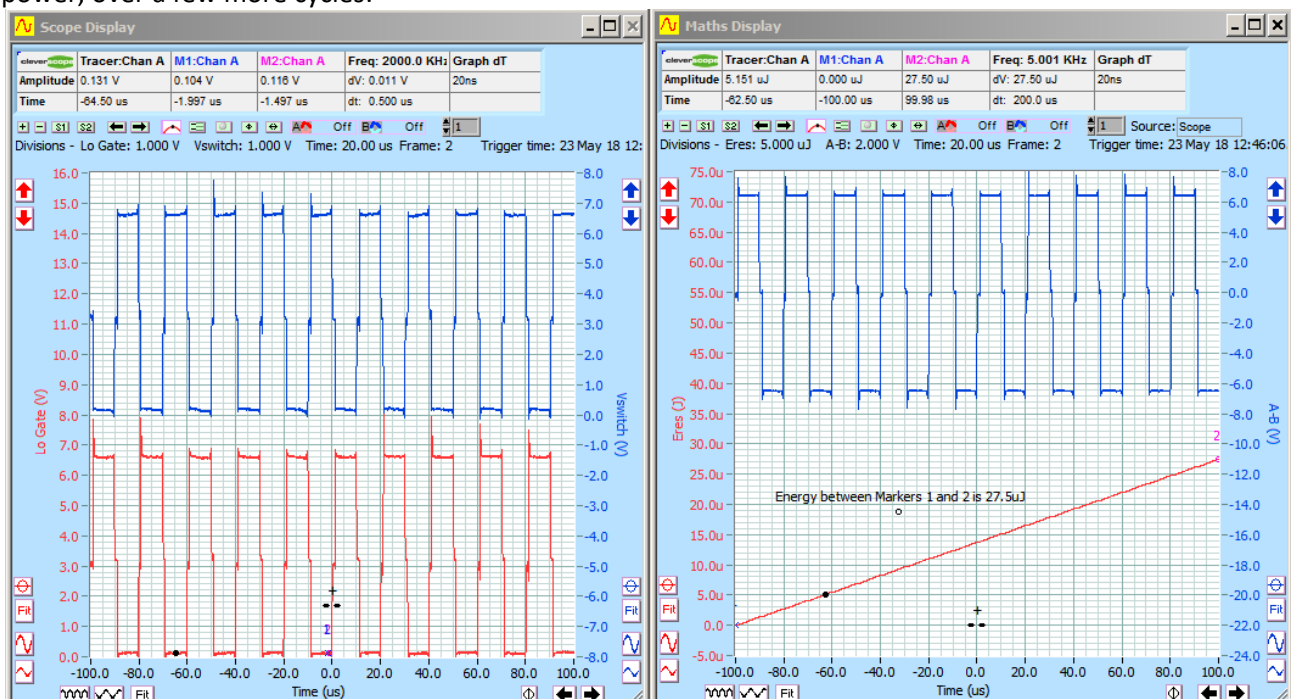
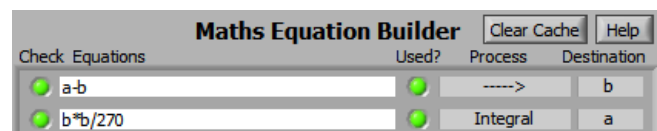
Maths Analyzer

The Maths Analyzer is used to derive information from the signal and display it, and optionally send it to Matlab. Here we have connected a 270 ohm resistor across the two half bridges, and used Maths Equation builder to display the difference between the two voltages on Channel B, and the derived instantaneous power being dissipated in the resistor in Chan A from V^2/R . We have labeled the axis accordingly:



You can see the peak power is about 150mW. The average power can be calculated by integrating the power to find energy, and then dividing by the time the integral was calculated over.

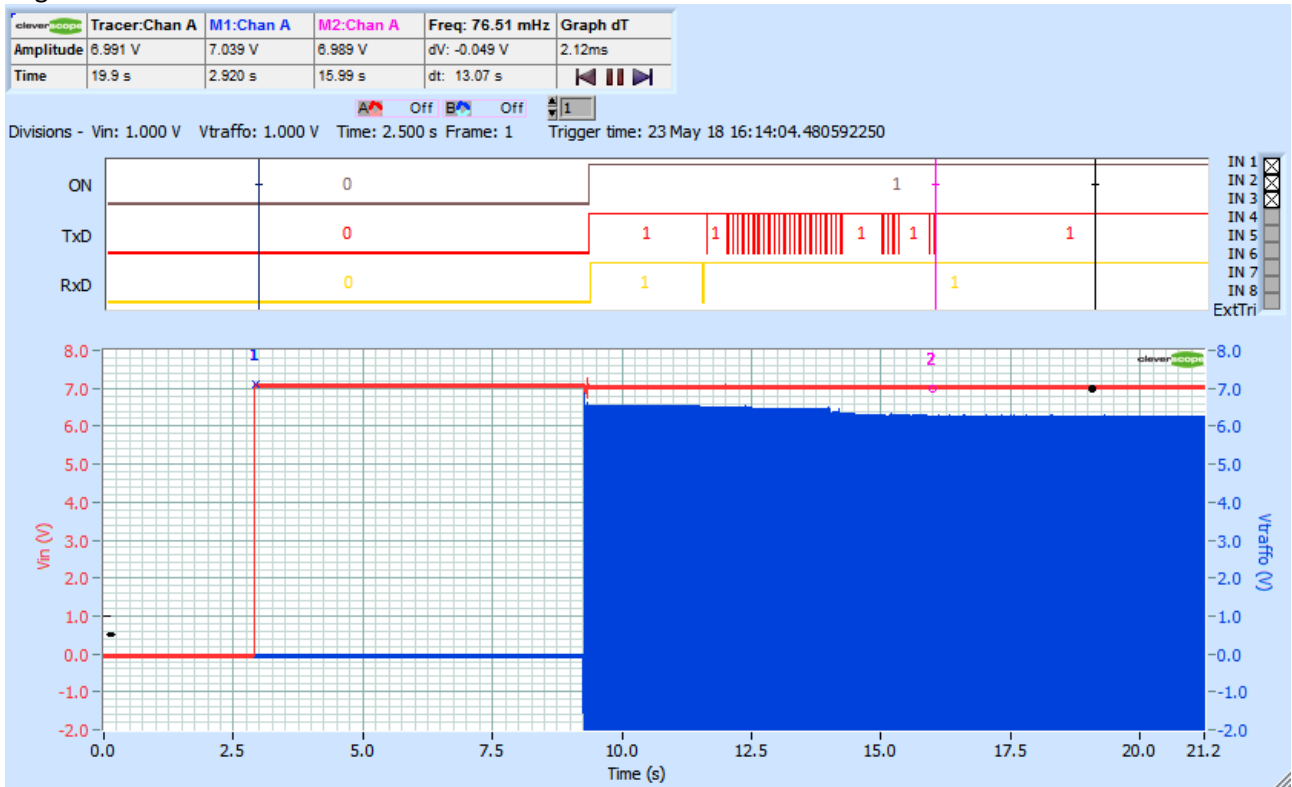
Here we redo the Maths, but now integrate the power, over a few more cycles:



The energy is 27.5 uJ over 100us. So the average power is $27.5 \text{ uJ} / 200 \text{ us} = 0.1375 \text{ W}$. We could do this over a longer time period to improve accuracy. Maths is useful wherever you want to derive further information.

Streaming Analyzer

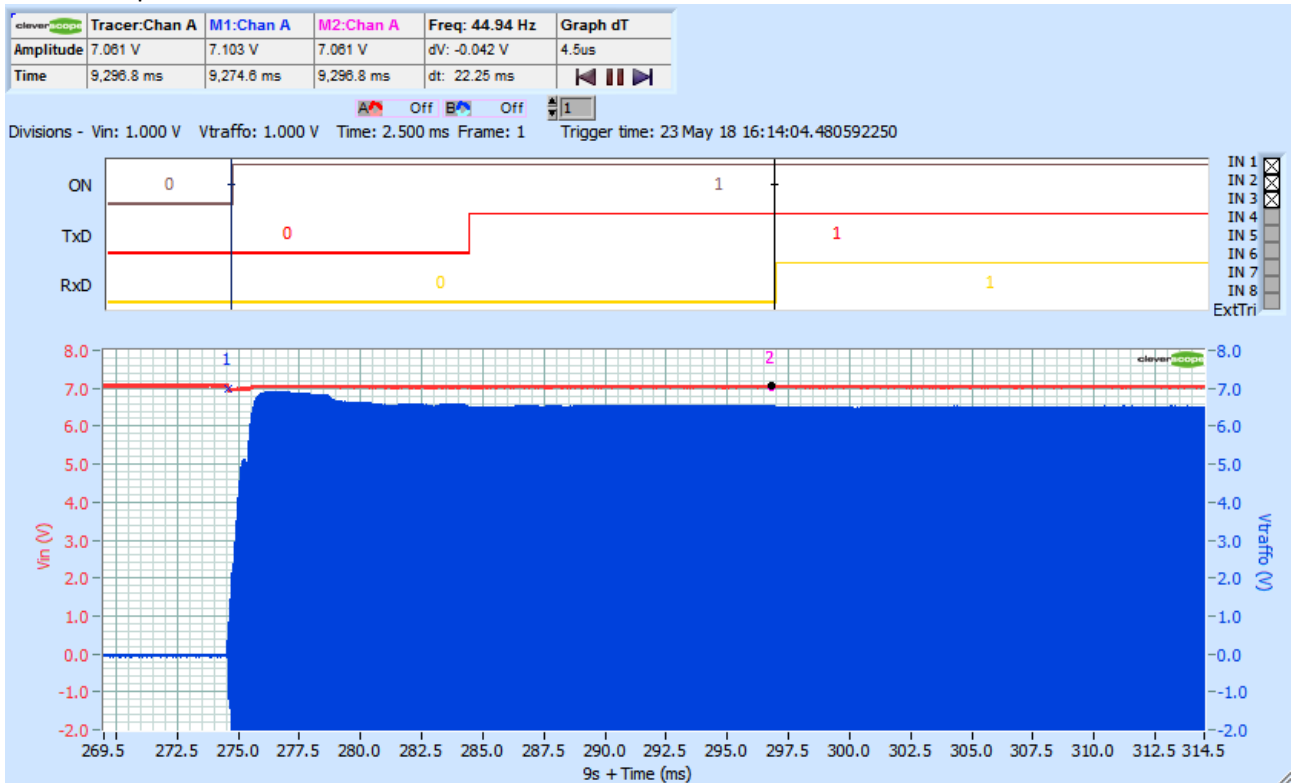
There are many situations where it is useful to see a sequence of actions of a longer timescale, and yet be able to drill down into the detail to understand what is happening. The Streaming Analyzer can acquire all channels (2 analog and 8 digital) at up to 3 Msample/sec, stream them to disk, and then quickly pan, zoom and analyze using the spectrum, maths and protocol decode analyzers. Here is the startup sequence for a single CS448 channel:



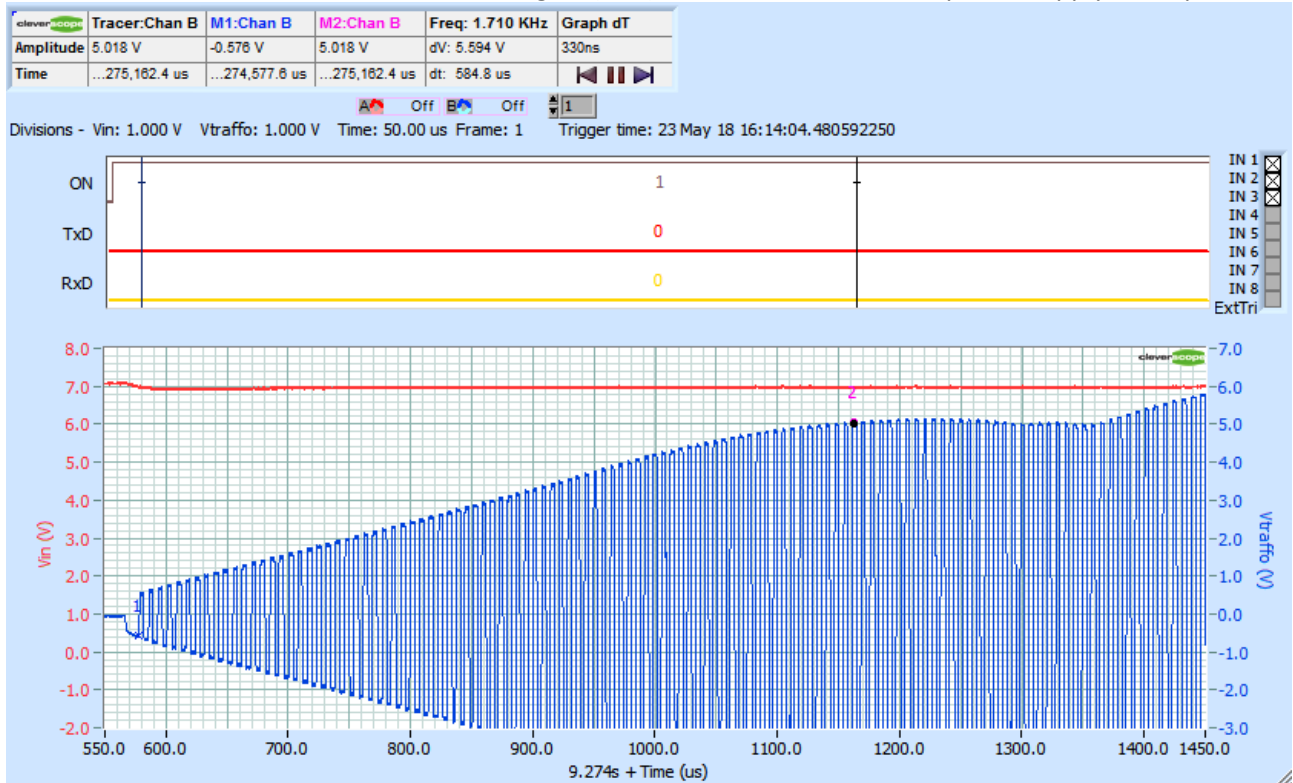
We captured 21.2 seconds at 3 MSPS. We are displaying both analog and digital signals.

Vin is a 7 V regulator that turns on at startup. Vtraffo is the isolated side power transformer coil. ON is a signal from the FPGA to turn on the Channel, TxD is the serial data transmitted from the FPGA, and RxD is the serial data received by the FPGA. Once we have captured the sequence we can examine portions of it. Using the Markers we see the total startup time is (M2-M1) or 13.07 seconds.

Here is the point at which ON is activated:

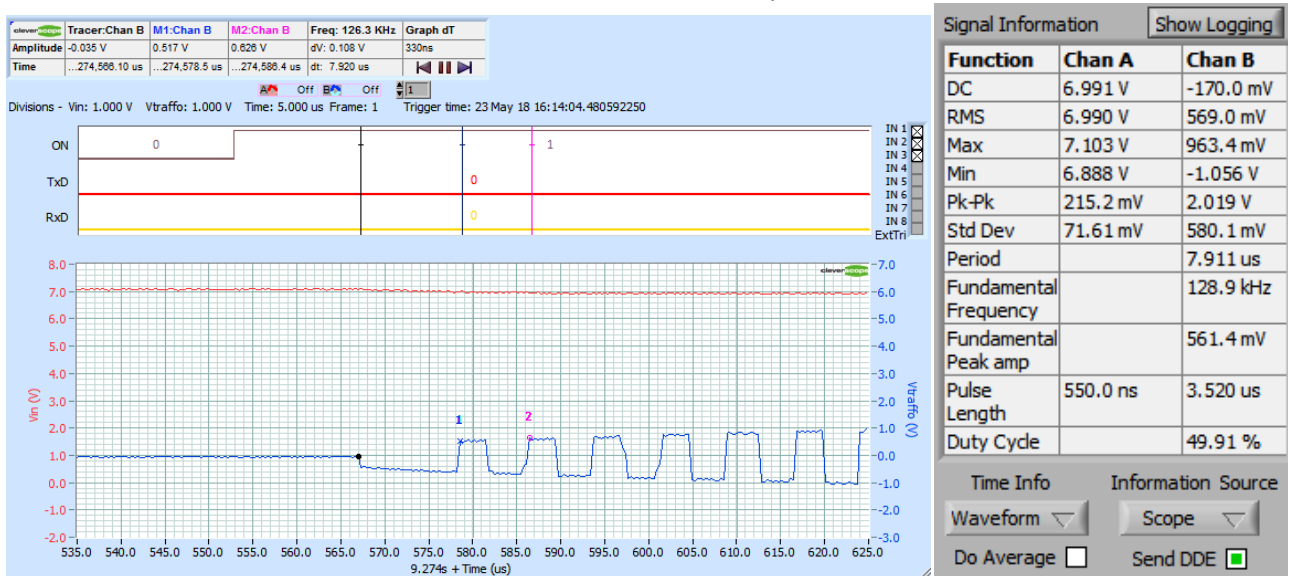


The isolated side turns on the UART return signal (RxD) 22.25 ms after ON. The power supply startup is:



The secondary switching signal settles at 5V after about 585 us (see marker dt), and then increases upwards again after about 200 us. The 7 V Vin loads down slightly as the switching starts.

With further zoom we can see the detail of the first few switch cycles:

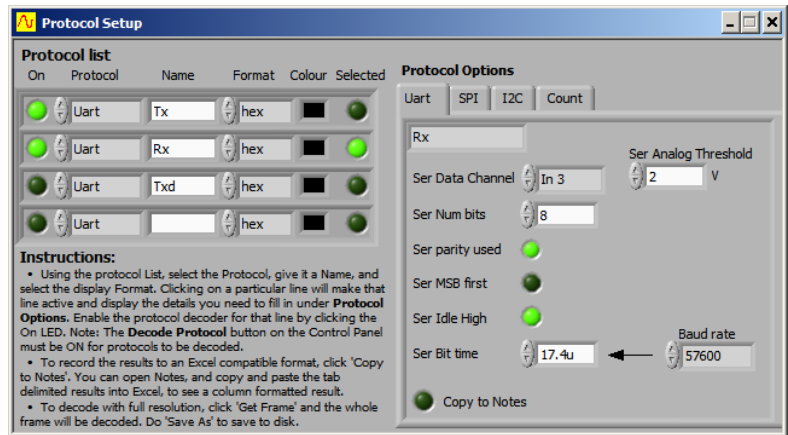


Graph dT is 330 ns (3 MHz sample rate). The switch frequency is about 126 kHz - signal information has a more accurate 128.9 kHz. We can see Vin (Chan A) is 6.991 V.

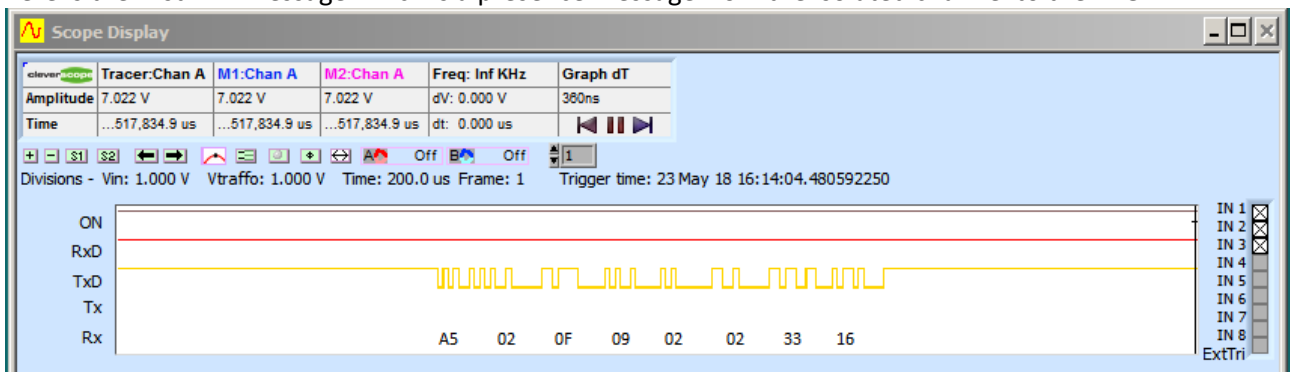
Protocol Analyzer

We can use the **Protocol Analyzer** to decode the serial data in the stream or a signal capture. Here is the protocol setup, with Rx shown:

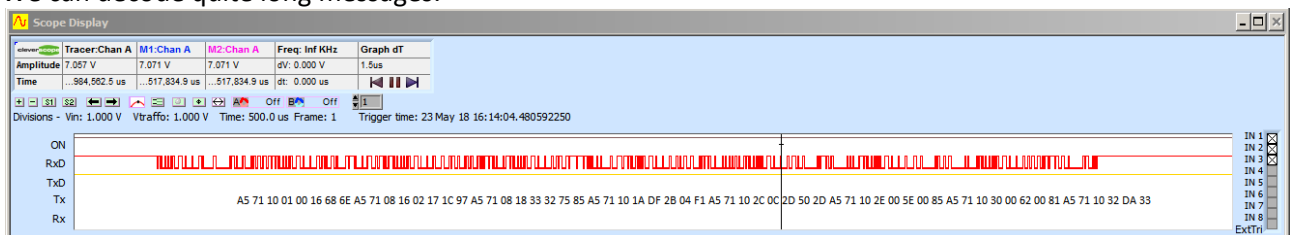
The names chosen are automatically placed on the digital portion of the scope or tracking display.



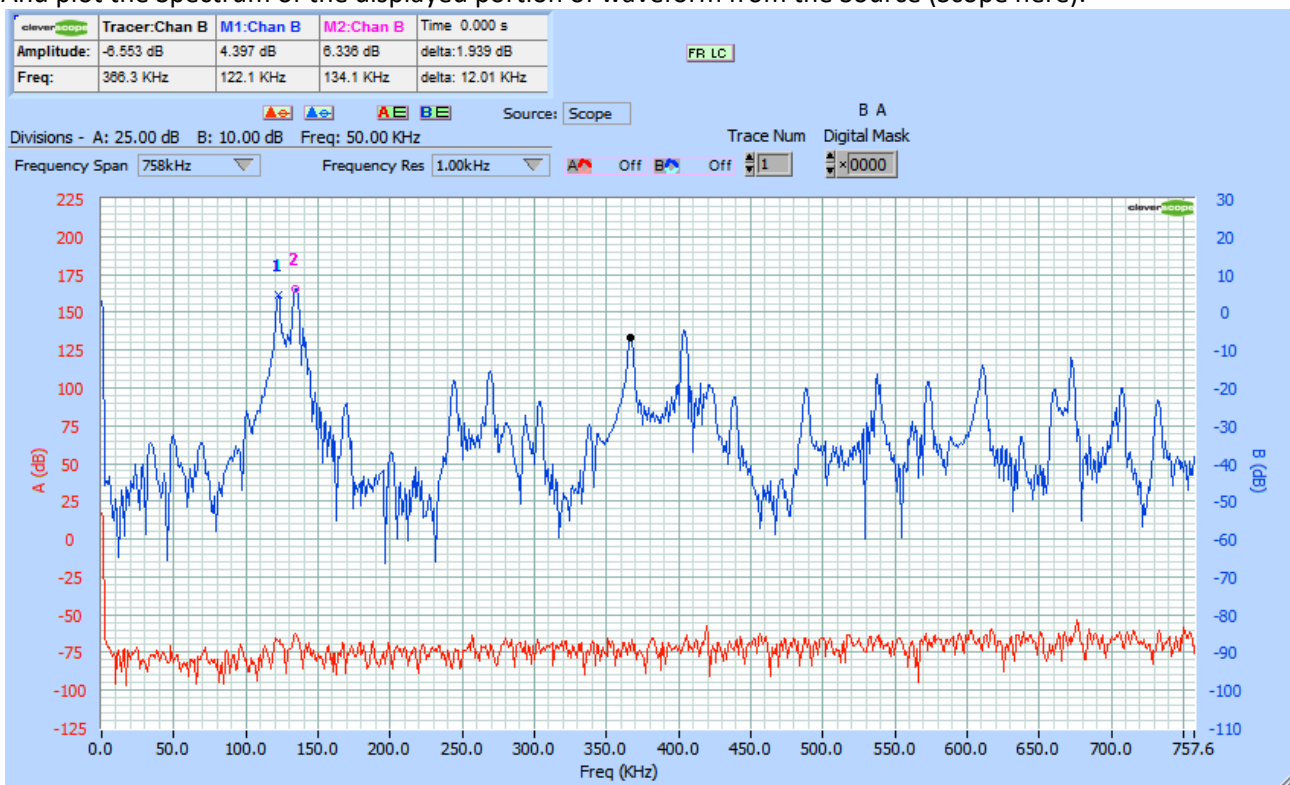
Here is the first RxD message which is a presence message from the isolated channel to the FPGA:



We can decode quite long messages:



And plot the Spectrum of the displayed portion of waveform from the Source (Scope here):



And we can see the spread spectrum modulated 122 - 134 kHz switching frequency.

Testimonials – unsolicited comments from users

"I am writing to let you know that the FRA has been working fine. I have played with it extensively and I have not found any problems, it has been very stable. Thank you for adding the LC button to the impedance trace where it displays the inductor and capacitance work very well. "

R Bolanos, USA

"I find your CS328A to be a very nice product, perhaps the best in its market segment. Of all the organizations I've ever worked with, as an engineer or otherwise, Cleverscope has the best support. "

M Linden, USA

"I can comment on the Cleverscope. I've been using it for about one year and it's simply great. Will probably buy another one next month. Linking two of them you have a four channel analog and 16 bit analyzer. "

E Brusque - Brazil

"Scope has been going well, bloody impressed with it."

H Rawnsley, New Zealand

"This CS328 is absolutely wonderful, I'm thrilled to bits with it. The software is light years in front of your opposition. The Maths function make me almost believe I'm using a LeCroy."

R Marmion, UK

Short Specification (A detailed technical specification and full manual are available on our website)

- Two 10, 12 or 14 bit analog channels sampling simultaneously at 100 MSa/sec. AC or DC coupled.
- 100 MHz Front end Analog bandwidth
- 4 or 8 MSamples of memory depth per channel
- USB 2.0 or Ethernet Interface to PC
- Optional 10MHz sine, square and triangle generator or 65MHz sine generator
- Gain automatically set from 20 mV full scale to 800 V full scale by choosing graph view and probe switch setting.
- Offset automatically set to correspond to the selected amplitude axis view. As an example, a graph with amplitude axis of 2.40 to 2.42 V automatically chooses a 20 mV range offset by +2.40 V – good for looking at small signals in single supply op-amp applications.
- Analog triggering of the waveform in view with a resolution of 1% of the display height. The analog trigger may optionally be conditioned with a low pass, high pass or noise filter.
- One external trigger, threshold adjustable from 0 to $\pm 20V$ in 40 mV increments.
- A dual hardware trigger system based on a rising or falling edge on any input signal, optionally qualified by a user defined digital input combination, triggering when one or both trigger conditions are met, and the time between triggers is less than, in or more than a user defined range, or the trigger count equals a user defined value.
- A rear panel I/O connector with a 100 Mbit/sec bi-directional LVDS/RS422 link, and three RS422 outputs defaulting to sampling started, trigger received and sampling stopped.
- Each channel (two analog, ext trigger and eight digital) includes 8M samples of storage, providing 40 ms of simultaneous storage for each channel, with 10 ns resolution.
- The sample storage may be allocated as between 2 to 3000 frames varying in size from 4 M to 2000 samples. These may be used as a history store for reviewing previously captured signals, or to capture up to 3000 trigger events with a minimal inter-frame delay, while maintaining time relative to the first trigger for all succeeding frames.
- 20 MHz 5th order Anti-alias filter for improved Spectrum Analysis performance.
- Input power range from 6 ~ 12V, 6W provided by a universal mains adaptor.
- Low jitter (1 ps rms) sampling clock for 70 dB spurious free dynamic range.
- Self calibration to ensure DC performance specifications are met.
- Optional external sampler clock interface

Visit our website for in-depth information, free demonstration software, training videos and examples. You can also send us an email or contact one of the distributors listed on the website.